

Problem Statement – Travel Planner Agent

Background

Planning a trip involves searching for destinations, transportation, accommodation, activities, weather conditions, maps, and costs. Managing all these factors manually is time-consuming and can lead to inefficient schedules, unexpected expenses, and poor travel decisions.

Problem

Traditional travel-planning methods require users to collect information from multiple sources and manually compare different options. They generally do not understand individual preferences or automatically adapt the itinerary when conditions such as weather, traffic, delays, availability, or budget change.

Proposed Challenge

The objective is to develop an **AI-powered Travel Planner Agent** that can understand a user's travel requirements, preferences, budget, and constraints and generate a personalized travel plan.

The agent should be able to:

- Suggest suitable destinations based on user preferences.
- Recommend transportation and accommodation options.
- Generate optimized day-wise itineraries.
- Consider budget, time, distance, and user interests.
- Integrate maps, weather, and local travel information.
- Dynamically modify the itinerary when conditions change.
- Provide alternative options when the original plan is unavailable.
- Reduce the manual effort involved in planning a complete trip.

Role of Agentic AI

Unlike a basic chatbot that only provides responses, the Travel Planner Agent should work toward a **specific travel goal**. It can understand the request, reason over available information, plan the trip, use relevant tools or APIs, evaluate options, and adapt the itinerary when required.

Technology Requirement

The solution must use **IBM Granite** and **IBM Cloud Lite services**. IBM Granite can support natural-language understanding, reasoning, and personalized itinerary generation, while IBM Cloud Lite provides the cloud environment and services required to build and demonstrate the application.

Expected Outcome

The final system should transform complex and manual travel planning into a **personalized, intelligent, budget-aware, and adaptive travel experience**, helping users make better travel decisions while saving time and effort.

ARCHITECTURE

